

KASHFLOW

AI-Powered SME Cash Flow Predictor Product Requirements Document

Document Version	1.0
Status	Draft — For Review
Date	21 May 2026
Product	Kashflow — SME Cashflow Predictor AI
URL	https://sme-cashflow-predictor-ai-513444368610.europe-west1.run.app/
Owner	Fasola Omogbolahan, Rita Ofuoku & Emmanuel Kpongo

1. Executive Summary

Kashflow is a cloud-native, AI-powered cash flow prediction platform purpose-built for Small and Medium-sized Enterprises (SMEs). It addresses one of the most critical pain points in small business management: unpredictable cash flow. By combining historical transaction analysis, machine-learning forecasting, and intelligent alerting, Kashflow empowers business owners and finance managers to anticipate liquidity shortfalls, plan working capital, and make confident financial decisions — without needing a CFO or a complex ERP system.

The platform is hosted on Google Cloud Run (europe-west1), ensuring high availability and auto-scaling without infrastructure overhead. It is designed to be lightweight, mobile-friendly, and actionable — turning raw financial data into plain-English insights.

2. Problem Statement

Cash flow mismanagement is the leading cause of SME failure globally. Despite having profitable businesses on paper, many SMEs are caught off-guard by timing mismatches between receivables and payables. The core challenges are:

- Lack of real-time visibility into upcoming cash positions
- Manual, spreadsheet-based forecasting that is error-prone and time-intensive
- No early-warning system for impending cash shortfalls
- Inability to model different financial scenarios (e.g. delayed invoice payment, large supplier bill)
- Fragmented financial data spread across banks, invoicing tools, and accounting software

Kashflow solves all of the above with an AI-driven, integrated, and automated approach tailored to the constraints and needs of SMEs.

3. Goals & Objectives

3.1 Product Goals

- Provide SMEs with accurate, AI-generated 30/60/90-day cash flow forecasts
- Reduce time spent on manual financial forecasting by at least 80%
- Alert users to cash shortfalls at least 14 days in advance
- Offer actionable recommendations alongside each forecast
- Integrate with widely-used accounting and banking platforms

3.2 Business Goals

- Achieve product-market fit with SMEs employing 1–250 staff

- Target markets: UK, EU, and English-speaking Africa (Nigeria, Kenya, Ghana)
- Reach 1,000 active businesses within 12 months of launch
- Maintain a Monthly Active User (MAU) retention rate above 65%

4. Target Users & Personas

Persona	Description	Primary Needs
The Bootstrapped Founder	Solo founder or micro-business owner, non-finance background, wears many hats	Simple visibility, no jargon, mobile-first, quick alerts
The SME Finance Manager	In-house finance lead at a 20–100 person company, familiar with spreadsheets	Detailed forecasts, scenario planning, exportable reports
The Accountant / Bookkeeper	External accountant managing 5–30 SME clients	Multi-client dashboard, white-label reports, API access
The Growth-Stage CEO	Scaling business, board-level reporting, investor communication needs	High-level summary, trend analysis, scenario modeling

5. Scope

5.1 In Scope — Version 1.0

- User authentication and onboarding (email + OAuth)
- Bank account connection via open banking (PSD2 / Open Banking API)
- AI-powered cash flow forecast for 30, 60, and 90 days
- Dashboard: cash position, forecast chart, upcoming inflows/outflows
- Proactive alerts for predicted shortfalls and anomalies
- Scenario planning: model the impact of delayed payments or large expenses
- PDF report export for each forecast period
- Mobile-responsive web application

5.2 Out of Scope — Version 1.0

- Native iOS / Android apps (planned for v2.0)
- Payroll processing or payroll integration
- Lending or credit product recommendations
- Multi-currency treasury management
- Full accounting/bookkeeping functionality

6. Functional Requirements

6.1 Onboarding & Authentication

- FR-01: Users must be able to register via email/password or OAuth (Google, Microsoft)
- FR-02: Email verification required before account activation
- FR-03: Guided onboarding flow (business name, industry, employee count, primary bank)
- FR-04: Multi-factor authentication (MFA) available and encouraged
- FR-05: Role-based access — Owner, Finance Manager, View-only

6.2 Data Integration

- FR-06: Connect one or more business bank accounts via Open Banking (PSD2)
- FR-07: Sync transaction history (minimum 12 months where available)
- FR-08: Optional integration with accounting platforms: Xero, QuickBooks, Sage, Wave
- FR-09: Manual data import via CSV upload for banks not supporting Open Banking
- FR-10: Automatic daily data refresh; manual refresh available on-demand

6.3 AI Cash Flow Forecasting Engine

- FR-11: Generate rolling 30/60/90-day cash flow forecasts on available data
- FR-12: Model must identify and weight: recurring income, recurring expenses, seasonal patterns, one-off transactions
- FR-13: Forecast must distinguish between committed (known) and predicted (AI-inferred) cash flows
- FR-14: Confidence intervals displayed on all forecast values (High / Medium / Low confidence)
- FR-15: Model must retrain/update automatically as new transaction data arrives
- FR-16: Plain-English summary of forecast generated automatically (e.g. 'You are likely to run low on 14 Jun — consider chasing Invoice #112')

6.4 Dashboard & Visualisation

- FR-17: Home dashboard displaying: current cash balance, forecast balance at 30/60/90 days, risk status (Green / Amber / Red)
- FR-18: Interactive time-series chart with actual vs. predicted cash position
- FR-19: Upcoming transactions panel (inflows and outflows for next 14 days)
- FR-20: Category breakdown of income and expenses (pie/bar chart)
- FR-21: Business health score card summarising liquidity, stability, and trend

6.5 Alerts & Notifications

- FR-22: Automated alert when forecast shows cash position dropping below user-defined threshold
- FR-23: Alert types: in-app notification, email, optional SMS/WhatsApp
- FR-24: Anomaly detection alert — unusual spikes or drops in daily transactions
- FR-25: Recurring payment reminders (e.g. 'VAT due in 7 days')
- FR-26: Users can configure alert sensitivity and thresholds in settings

6.6 Scenario Planning

- FR-27: Scenario builder allowing users to add hypothetical inflows/outflows to the forecast
- FR-28: Side-by-side comparison of base forecast vs. scenario forecast
- FR-29: Pre-built scenario templates: 'Client pays 30 days late', 'Hire 2 staff', 'Large equipment purchase'
- FR-30: Save and name up to 5 scenarios per account

6.7 Reporting & Export

- FR-31: Generate branded PDF cash flow reports for any selected date range
- FR-32: Report must include: executive summary, forecast chart, category breakdown, scenario comparison (if saved)
- FR-33: Export underlying data to CSV/Excel
- FR-34: Accountant role can generate reports across all managed client accounts

6.8 Settings & Administration

- FR-35: Manage connected bank accounts and integrations
- FR-36: Invite and manage team members with role assignment
- FR-37: Billing and subscription management
- FR-38: Audit log of AI forecast generation and data syncs

7. Non-Functional Requirements

Category	Metric / Target	Detail
Performance	< 2s page load	Dashboard and forecast pages must load within 2 seconds on standard broadband
Performance	< 30s forecast	Full AI forecast generation must complete within 30 seconds of data refresh
Availability	99.9% uptime SLA	Hosted on GCP Cloud Run with auto-scaling; health checks every 60 seconds

Security	SOC 2 alignment	Data encrypted at rest (AES-256) and in transit (TLS 1.3); no plain-text storage of banking credentials
Security	PSD2 compliance	Open Banking integrations must comply with PSD2 and FCA guidelines where applicable
Privacy	GDPR / NDPR	Compliant with UK GDPR, EU GDPR, and Nigeria Data Protection Regulation
Scalability	10,000 concurrent users	Cloud Run auto-scales horizontally; target support for 10k concurrent sessions without degradation
Accessibility	WCAG 2.1 AA	Minimum AA conformance for all user-facing screens
Browser Support	Last 2 major versions	Chrome, Firefox, Safari, Edge; mobile Chrome and Safari on iOS/Android

8. Technical Architecture

8.1 Infrastructure

- Hosting: Google Cloud Run (europe-west1) — fully managed, containerised, serverless
- CI/CD: GitHub Actions → Docker image build → Cloud Run deployment
- Secrets: Google Secret Manager
- Database: Cloud SQL (PostgreSQL) for structured data; Cloud Firestore for real-time events
- File storage: Google Cloud Storage (GCS) for generated PDF reports and CSV exports

8.2 AI / ML Stack

- Forecasting model: Time-series ensemble (ARIMA + LSTM neural network) trained per business
- Feature inputs: Daily transaction aggregates, day-of-week, seasonal indices, business category
- Model serving: Vertex AI Endpoints for inference; retrained weekly per account
- NLP summaries: Claude API (Anthropic) for plain-English forecast narrative generation

8.3 Integrations

- Open Banking: TrueLayer / Plaid for UK/EU bank connections
- Accounting: Xero API, QuickBooks API, Sage Business Cloud API
- Notifications: SendGrid (email), Twilio (SMS), WhatsApp Business API
- Payments: Stripe for subscription billing

9. User Stories

ID	User Story	Acceptance Criteria
US-01	As a founder, I want to see my cash balance today and in 30 days so I know if I can pay my staff next month.	Dashboard shows current balance + 30-day forecast; confidence level visible
US-02	As a finance manager, I want to model the impact of a client paying 45 days late so I can plan a credit line in advance.	Scenario builder adjusts forecast and shows adjusted balance timeline
US-03	As a business owner, I want to receive a WhatsApp alert when my cash is about to fall below £5,000 so I can act quickly.	Alert fires 14+ days in advance; threshold configurable; delivery confirmed
US-04	As an accountant, I want to generate a branded PDF cash flow report for my client so I can present it at their monthly review.	Report generates in < 60 seconds; includes logo, forecast chart, summary
US-05	As a CEO, I want the forecast to tell me in plain English what my biggest financial risk is this quarter.	AI narrative appears on dashboard; updated with each sync cycle
US-06	As a user, I want to connect my business bank account in under 5 minutes without sharing my login credentials.	OAuth-based Open Banking flow; connection confirmed < 5 min; no credentials stored

10. Feature Prioritisation (MoSCoW)

Feature	Priority	Release
User authentication & onboarding	Must Have	v1.0
Bank account connection (Open Banking)	Must Have	v1.0
AI 30/60/90-day cash flow forecast	Must Have	v1.0
Dashboard with cash position & forecast chart	Must Have	v1.0
Shortfall alerts (email)	Must Have	v1.0
PDF report export	Must Have	v1.0
Scenario planning builder	Should Have	v1.0
WhatsApp / SMS alerts	Should Have	v1.0
Xero / QuickBooks integration	Should Have	v1.1
Multi-user / team roles	Should Have	v1.1
Accountant multi-client dashboard	Could Have	v1.2
Anomaly detection alerts	Could Have	v1.2
Business health score card	Could Have	v1.2

Native mobile app (iOS/Android)	Won't Have	v2.0
Multi-currency treasury management	Won't Have	v2.0

11. UX & Design Principles

- Clarity over complexity: every screen must communicate one primary insight
- Progressive disclosure: show simple summary first; detail on demand
- Mobile-first responsive design: all core workflows must be completable on a smartphone
- Trust through transparency: always show data freshness, confidence levels, and data sources
- Actionable insights: every forecast must link to a suggested next action
- Accessibility: WCAG 2.1 AA minimum; support screen readers, keyboard navigation, sufficient colour contrast

12. Success Metrics & KPIs

Metric	Target (12 months)	Measurement Method
Active business accounts	1,000	Database account count
Monthly Active Users (MAU)	> 65% retention	Login events / analytics
Forecast accuracy	> 85% within 10% margin	Predicted vs. actual cash at 30 days
Onboarding completion rate	> 70%	Funnel analytics (bank connected)
Alert false positive rate	< 10%	User-dismissed alerts marked as incorrect
NPS score	> 45	In-app quarterly NPS survey
Page load time (P95)	< 2 seconds	Google Cloud Monitoring / Lighthouse

13. Risks & Mitigations

Risk	Severity	Mitigation
Open Banking API changes or partner outages	High	Multi-provider strategy (TrueLayer + Plaid); manual CSV fallback always available

AI forecast accuracy below expectation for new accounts with < 3 months data	High	Display confidence as 'Low' for data-sparse accounts; improve model with more data over time; clear user communication
Regulatory compliance (PSD2, GDPR, NDPR)	High	Legal review before launch; DPA in place; FCA registration assessment; GDPR DPO appointed
Low user trust in AI-generated financial predictions	Medium	Transparency about methodology; confidence intervals; accuracy track record shown over time
SME churn due to low engagement after onboarding	Medium	Weekly digest emails; proactive alerts; frictionless return experience
Data breach or security incident	Critical	No credential storage; end-to-end encryption; penetration testing pre-launch; incident response plan

14. Release Roadmap

Release	Target Date	Key Deliverables
Alpha	Q3 2026	Auth, bank connection, basic 30-day forecast, dashboard MVP — closed beta with 20 SMEs
v1.0	Q4 2026	Full 30/60/90-day AI forecast, alerts (email), scenario builder, PDF export, public launch
v1.1	Q1 2027	Xero/QuickBooks integration, WhatsApp/SMS alerts, multi-user roles, anomaly detection
v1.2	Q2 2027	Accountant multi-client portal, business health score, expanded Open Banking (EU/Africa)
v2.0	Q3 2027	Native iOS & Android apps, multi-currency support, embedded lending referrals

15. Open Questions & Dependencies

Open Questions

- Will the product launch under the 'Kashflow' brand name, or is rebranding required given existing KashFlow (IRIS) trademark?
- What is the primary pricing model — freemium, subscription, usage-based, or accountant-led reseller?
- Which Open Banking provider takes priority for African markets (TrueLayer does not cover Nigeria/Kenya)?
- Is there a preferred AI/ML framework for the forecasting engine, or is this an open decision?
- What is the minimum viable historical data threshold before a forecast is shown to users?

Dependencies

- Open Banking / PSD2 API partner agreement (TrueLayer or equivalent)
- Anthropic API access for narrative generation
- Stripe account setup for subscription billing
- Legal review: FCA registration (UK), GDPR Data Processing Agreement, NDPR compliance (Nigeria)
- Design system / brand guide to be finalised before Alpha build

This document was prepared on 21 May 2026 and reflects current product understanding. It should be reviewed and updated at each sprint planning cycle.